POWER QUERY

Complete Training Notes and Practical Revision

1. What Is Power Query?

Power Query is an Excel tool used to import, clean, transform, combine, and prepare data for analysis.

Its main advantage is that it saves transformation steps. When the source data is updated, the saved steps can be applied again by refreshing the query.

Business example: A company receives a new sales file every day. Instead of cleaning the file manually every day, we can build a Power Query workflow once and refresh it when the source data changes.

2. Importing Data

Common ways to import data into Power Query:

- Data → From Table/Range: Import data from an Excel table or selected range.
- Data → From Text/CSV: Import a CSV or text file.
- Data → From Folder: Combine multiple similar files from a folder.

Business logic: Choose the import method based on where the data comes from and whether new files will arrive regularly.

3. Transforming Data

Common transformation tasks:

- Change Data Type: Set columns to the correct type.
- Rename Columns: Give columns clear, meaningful names.
- Split Column: Separate combined information using a delimiter.
- Filter Rows: Keep only the records needed for analysis.
- Trim: Remove unnecessary leading and trailing spaces.

Examples:

- Split “Rahul,India” using a comma.
- Filter Country to show Nigeria only.
- Trim “ India ” so it becomes “India”.

4. Data Types

Choose the correct data type for every column.

- TransactionID: Text

- Date: Date
- Country: Text
- Salesperson: Text
- Quantity: Whole Number
- UnitPrice: Decimal Number or Currency, depending on the data and reporting requirements.

Important: A transaction ID may contain digits, but it is an identifier, not a quantity to be calculated.

5. Cleaning Missing Values

A blank value does not always mean that the entire record is invalid.

- Blank Salesperson: Investigate whether the name can be recovered.
- Blank Quantity: Investigate the source because the missing value affects calculations.
- Entirely blank row: Remove it after confirming that all fields are empty.

Never automatically replace every blank with zero. Zero and missing information have different business meanings.

6. Handling Invalid Values

Example: Quantity contains “ten” instead of a number.

Recommended process:

1. Inspect the invalid value.
2. Check the original source.
3. Correct it if the intended value is verified.
4. Set the column to Whole Number.
5. Check for conversion errors.

Do not guess or invent values.

7. Removing Duplicates

Before removing duplicates:

1. Understand the data grain — what one row represents.
2. Identify the columns that define a unique record.
3. Compare repeated records.
4. Verify whether the repetition is accidental.
5. Remove confirmed duplicates and validate the resulting totals.

Important: A repeated TransactionID may represent multiple product lines. Do not remove rows based only on a repeated ID unless the data structure confirms that this is appropriate.

8. Final Data Quality Check

Before loading the cleaned data, check:

- Column names and data types
- Extra spaces
- Missing values
- Invalid values and conversion errors
- Duplicate records
- Completely blank rows
- Row counts and key totals before and after cleaning

Investigate unexpected changes in totals or row counts.

9. Loading Data into Excel

After validating the query:

1. Select Home → Close & Load.
2. Check the output in Excel.
3. Confirm the correct columns and records were loaded.
4. Review the Queries & Connections pane if you need to inspect the query.

10. Refreshing Data

When the source data changes:

1. Update the connected source.
2. Return to Excel.
3. Select Data → Refresh All.
4. Wait for the query to finish.
5. Validate the refreshed output.

A query connected to a specific file normally refreshes that source. A properly configured folder query can process eligible files added to the folder.

11. Complete Power Query Workflow

IMPORT → TRANSFORM → CLEAN → VALIDATE → LOAD → REFRESH

- Import: Bring data into Power Query.
- Transform: Change the structure and format.
- Clean: Resolve data quality issues.
- Validate: Check data types, errors, duplicates, and totals.
- Load: Send the prepared data to Excel.
- Refresh: Rerun the saved steps when the source is updated.

12. Key Business Lessons

1. Correct data types are essential for accurate calculations.
2. Standardized text prevents the same country or salesperson from appearing under different names.
3. Missing values should be investigated rather than blindly deleted.
4. Invalid values should be corrected only when verified.
5. Duplicate records should be checked against the data grain.
6. Completely blank rows can be removed after confirmation.
7. Refresh automates repeated transformations but does not replace quality checks.

Final Revision Statement

Power Query helps create a repeatable data preparation process. Instead of manually cleaning every new file, we define the transformation steps once, validate the output, load it into Excel, and refresh it when the source data changes.